

BÁO CÁO BÀI TẬP VỀ NHÀ NGÀY 19/05/2026

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1. Bài 3.01 – file_server
 - Chạy server ở cổng 8888

The screenshot shows a Visual Studio Code editor with a C program named `file_server.c` open. The program is a file server that listens on port 8888. It uses `signal(SIGCHLD, signalHandler);` to handle child processes. The `main` function sets up a `listen_sock` and enters a `while (1)` loop to accept connections. When a client connects, it forks a child process to handle the request. The child process reads data from the client, processes it, and writes it to a file. The parent process then prints the client ID and continues to accept more connections.

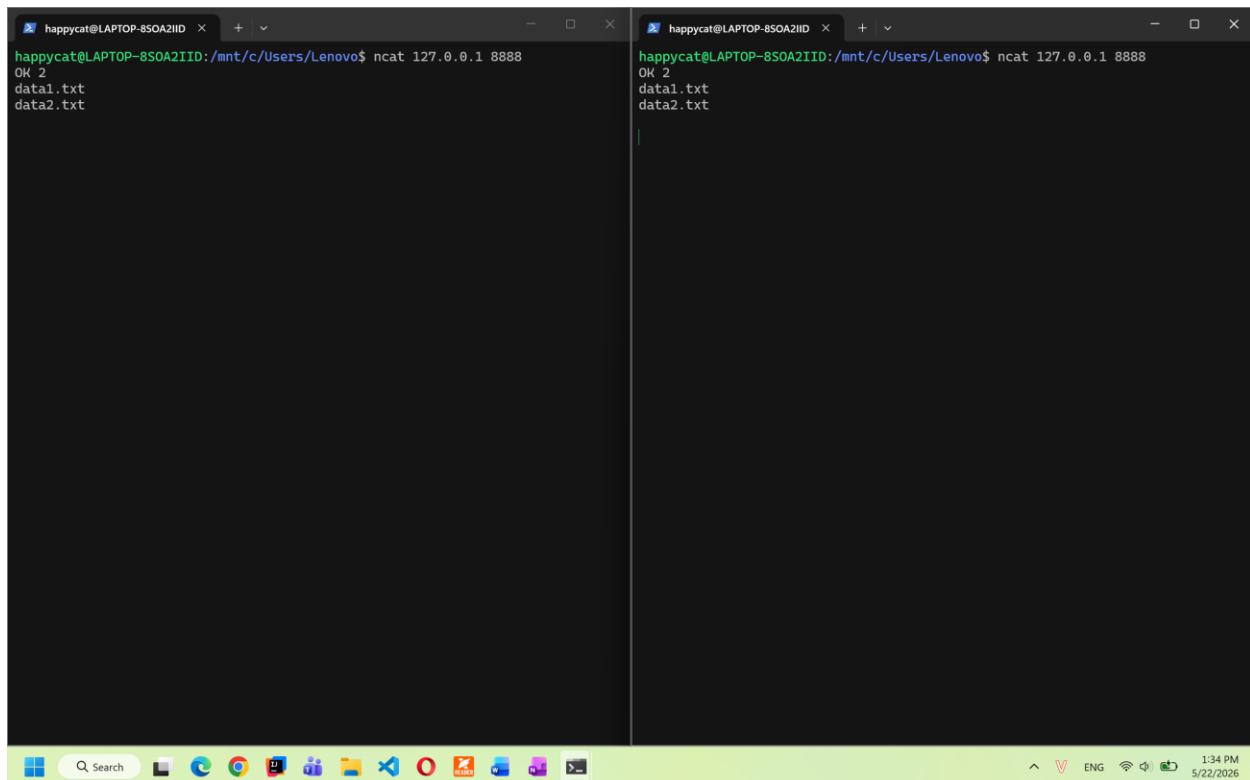
```
55 signal(SIGCHLD, signalHandler);
56 while (1) {
57     // neu co yeu cau ket noi
58     // chap nhan va ghi vao mang clients
59     int client = accept(listen_sock, NULL, NULL);
60     printf("New client accepted: %d\n", client);
61
62     if (fork() == 0) {
63         close(listen_sock); // dong listen_sock vi khong dung
64         signal(SIGCHLD, SIG_DFL); // tien trinh con khong quan tam SIGCHLD
65
66         char buffer[63686];
67         char filepath[1024];
68         char temp_file_list[60000] = "";
```

The terminal window shows the command to compile and run the program:

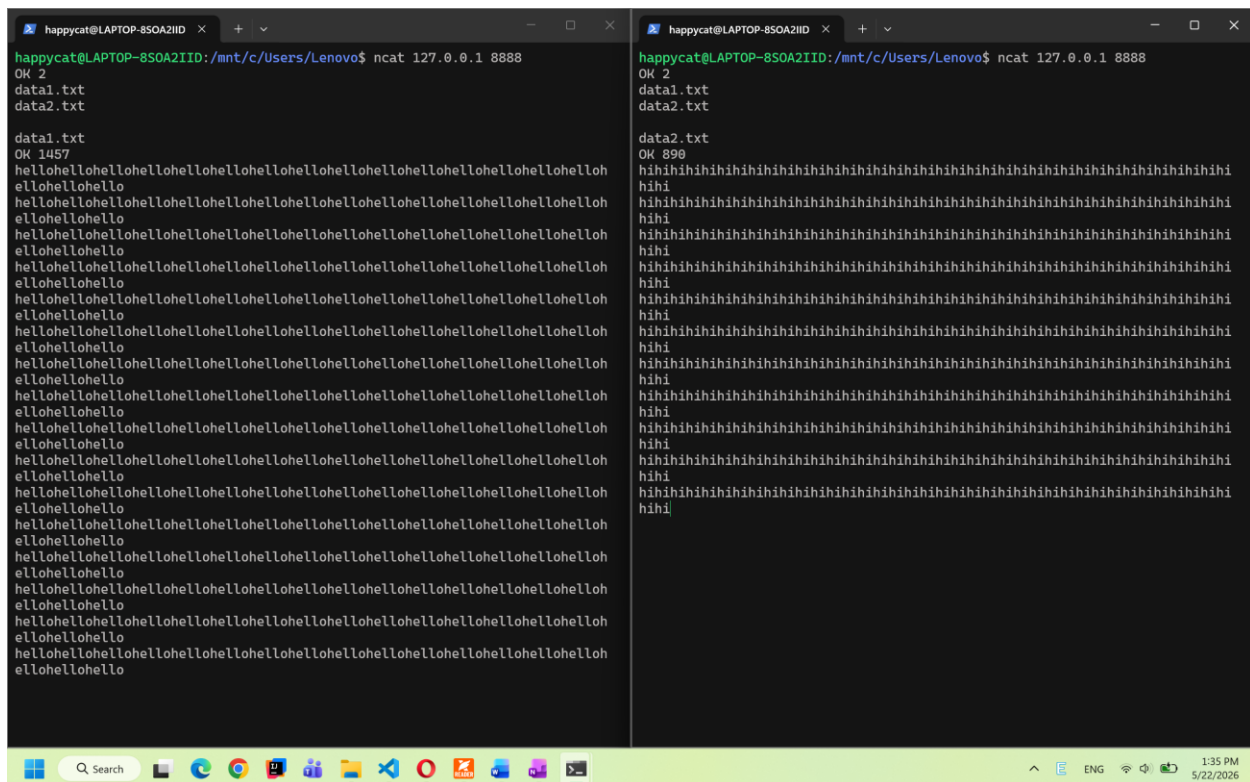
```
happy@LAPTOP-8SOA2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc file_server.c -o build/file_server && ./build/file_server
Server is listening at port 8888...
```

A notification at the bottom right states: "Your application running on port 8888 is available. See all forwarded ports".

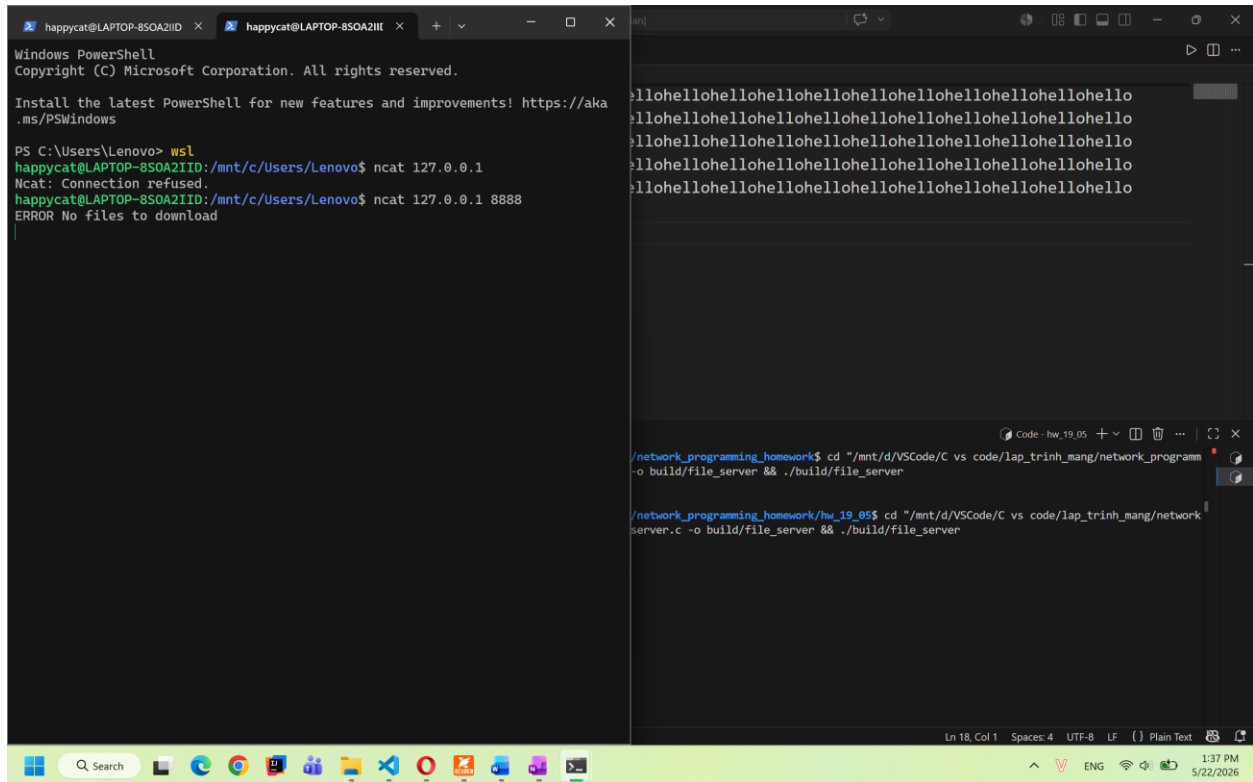
- Tạo 2 client kết nối với server, server sẽ gửi OK số_file danh_sách_file



- Client gửi tên file, server sẽ gửi OK dung_lượng_file nội_dung và đóng kết nối



- Trường hợp không có file (trong thư mục ./data)



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

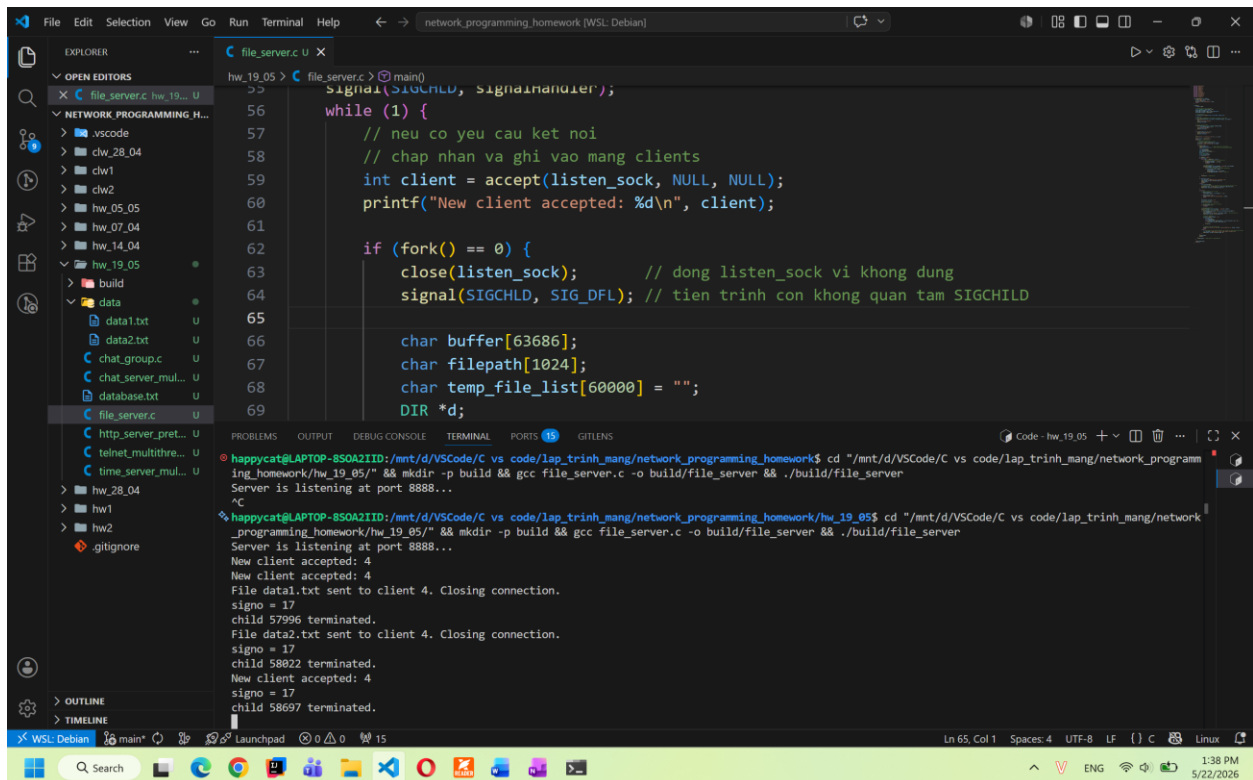
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Lenovo> wsl
happyocat@LAPTOP-8SOA2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1
Ncat: Connection refused.
happyocat@LAPTOP-8SOA2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
ERROR No files to download

/network_programming_homework$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05" && ./build/file_server

/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05" && ./build/file_server
```

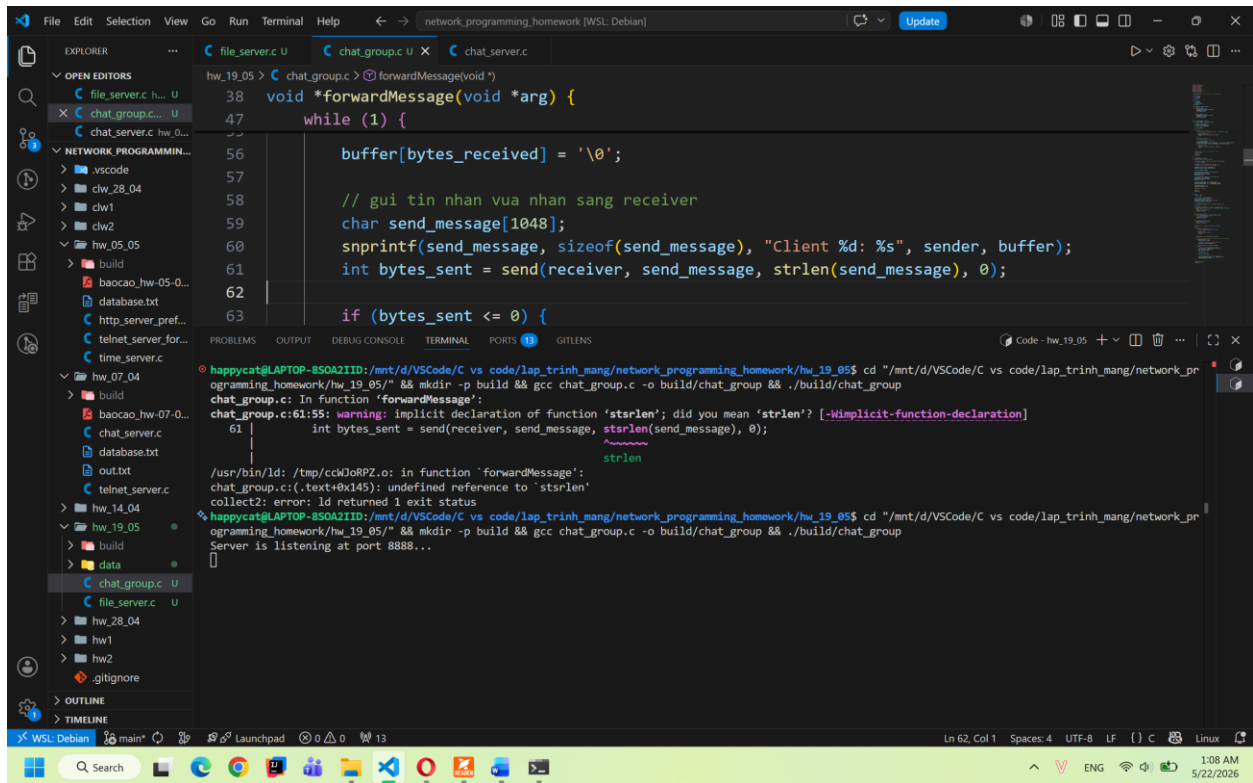
- Kết quả phía server



```
file_server.c
56 signal(SIGCHLD, signal_handler);
57 while (1) {
58     // neu co yeu cau ket noi
59     // chap nhan va ghi vao mang clients
60     int client = accept(listen_sock, NULL, NULL);
61     printf("New client accepted: %d\n", client);
62
63     if (fork() == 0) {
64         close(listen_sock); // dong listen_sock vi khong dung
65         signal(SIGCHLD, SIG_DFL); // tien trinh con khong quan tam SIGCHLD
66
67         char buffer[63686];
68         char filepath[1024];
69         char temp_file_list[60000] = "";
70         DIR *d;
71
72         // ... (rest of the code) ...
73     }
74 }
```

```
happyocat@LAPTOP-8SOA2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05" && mkdir -p build && gcc file_server.c -o build/file_server && ./build/file_server
Server is listening at port 8888...
New client accepted: 4
New client accepted: 4
File data1.txt sent to client 4. Closing connection.
child 57996 terminated.
File data2.txt sent to client 4. Closing connection.
child 58022 terminated.
New client accepted: 4
child 58097 terminated.
```

2. Bài 3.02 – chat_group
- Chạy server ở cổng 8888



```
void *forwardMessage(void *arg) {
    while (1) {
        buffer[bytes_received] = '\0';

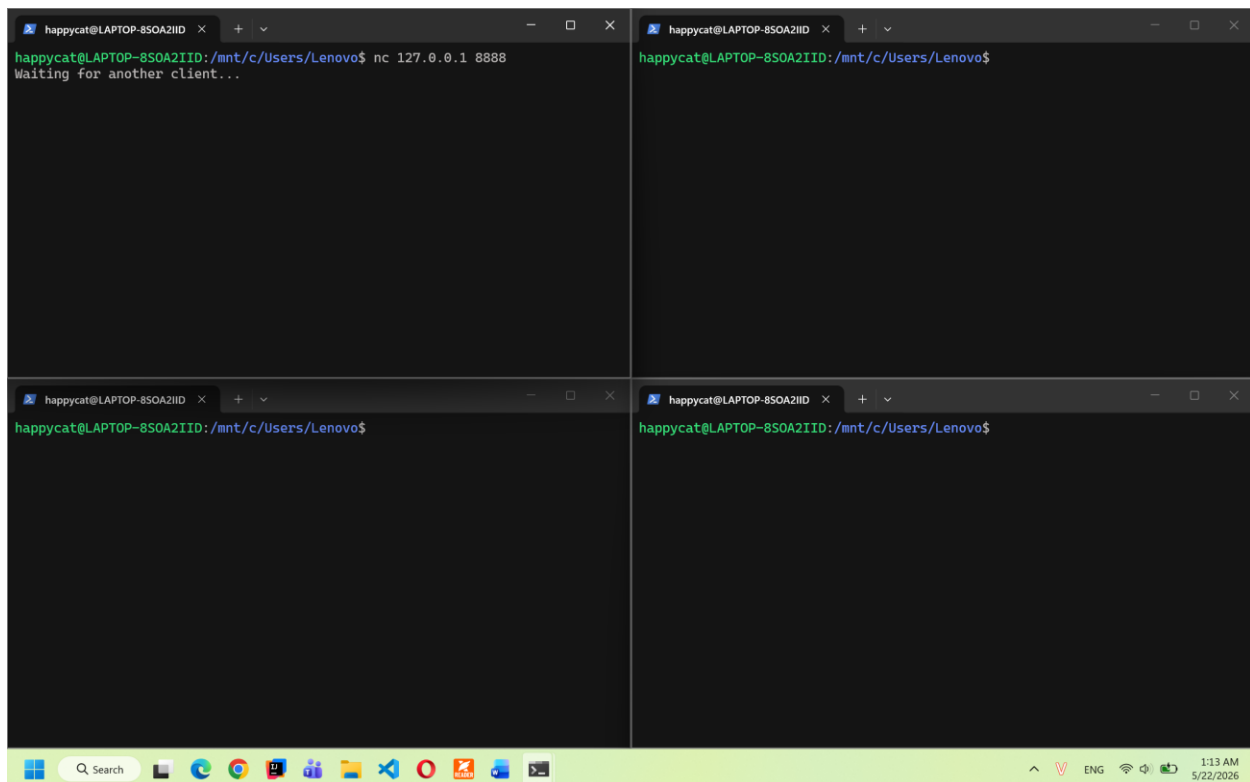
        // gui tin nhan vua nhan sang receiver
        char send_message[1024];
        snprintf(send_message, sizeof(send_message), "Client %d: %s", sender, buffer);
        int bytes_sent = send(receiver, send_message, strlen(send_message), 0);

        if (bytes_sent <= 0) {
```

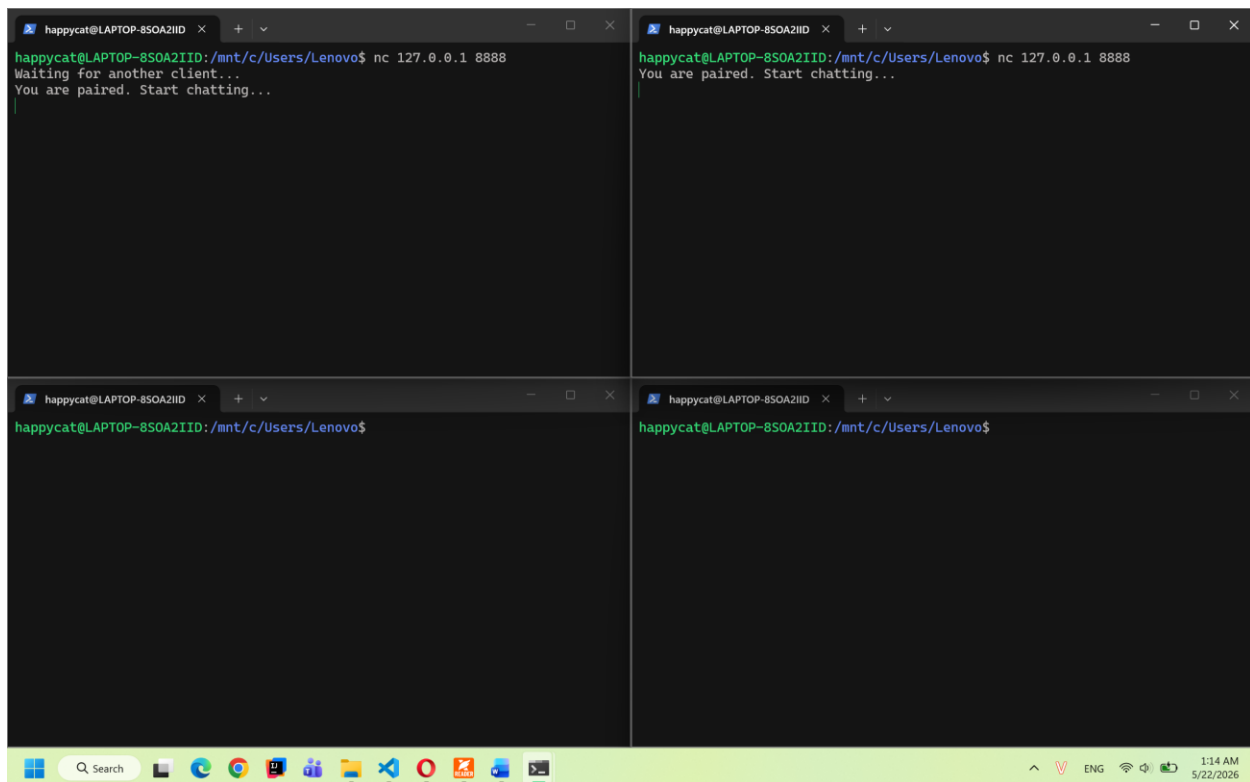
```
chat_group.c: In function 'forwardMessage':
chat_group.c:61:55: warning: implicit declaration of function 'strlen'; did you mean 'strlen'? [-Wimplicit-function-declaration]
        int bytes_sent = send(receiver, send_message, strlen(send_message), 0);
                                                         ^~~~~~
                                                         strlen
/usr/bin/ld: /tmp/ccJ0rPZ.o: in function 'forwardMessage':
chat_group.c:(.text+0x145): undefined reference to 'strlen'
collect2: error: ld returned 1 exit status

happycat@LAPTOP-850A21TD:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_group.c -o build/chat_group && ./build/chat_group
Server is listening at port 8888...
```

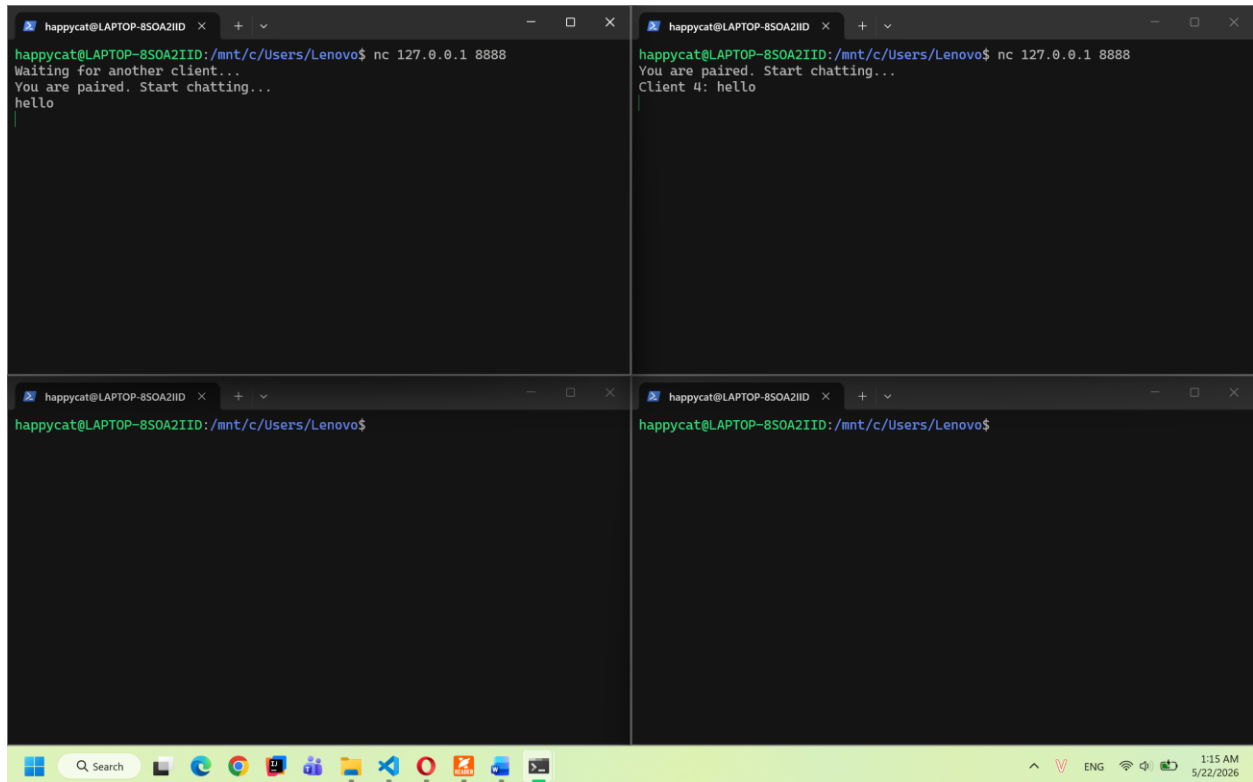
- Kết nối 1 client vào server, server sẽ gửi thông báo “Waiting for another client...”



- Kết nối client 2 vào server



- Client 1 gửi tin nhắn (fd là 4), client 2 nhận được và hiển thị



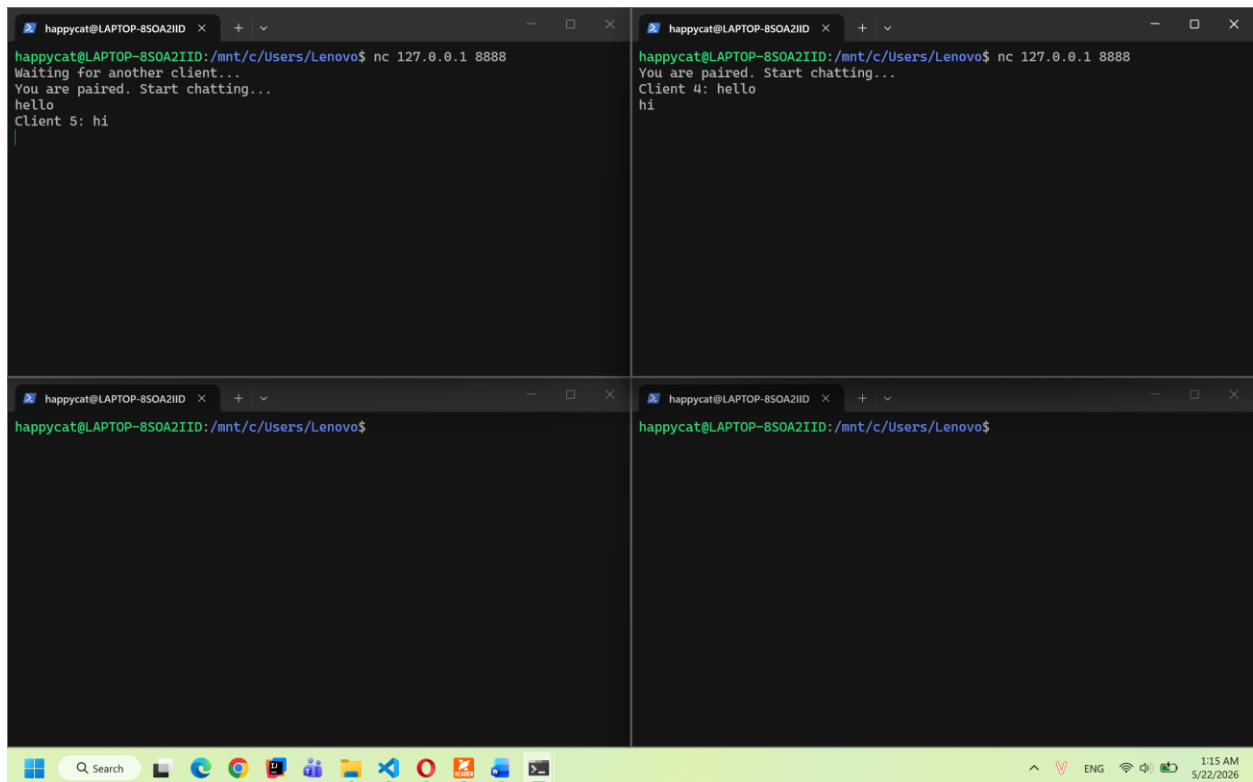
```
happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Waiting for another client...
You are paired. Start chatting...
hello

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
You are paired. Start chatting...
Client 4: hello

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$
```

- Tương tự với client 2 (fd là 5)



```
happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Waiting for another client...
You are paired. Start chatting...
hello
Client 5: hi

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
You are paired. Start chatting...
Client 4: hello
hi

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$

happy cat@LAPTOP-8S0A2IID: /mnt/c/Users/Lenovo$
```

- Thực hiện kết nối và nhắn tin tương tự với client 3, 4

The image shows four terminal windows running the 'happycat' chat server. Each window displays the server's state as it receives connections and messages from different clients.

- Top-left window:** Shows Client 5 sending 'hi'.
- Top-right window:** Shows Client 4 sending 'hello' and 'hi'.
- Bottom-left window:** Shows Client 7 sending 'xin chào', 'hiiii', and 'chào bạn'.
- Bottom-right window:** Shows Client 6 sending 'xin chào', 'hiiii', and 'chào bạn'.

- Kết quả server

The image shows the VS Code editor with the 'chat_group.c' source code and the terminal output of the chat server.

Source Code (chat_group.c):

```
38 void *forwardMessage(void *arg) {
47     while (1) {
56         buffer[bytes_received] = '\0';
57
58         // gửi tin nhắn vừa nhận sang receiver
59         char send_message[1024];
60         snprintf(send_message, sizeof(send_message), "Client %d: %s", sender, buffer);
61         int bytes_sent = send(receiver, send_message, strlen(send_message), 0);
62
63         if (bytes_sent <= 0) {
```

Terminal Output:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_group.c -o build/chat_group && ./build/chat_group
chat_group.c: In function 'forwardMessage':
chat_group.c:61:55: warning: implicit declaration of function 'strlen'; did you mean 'strlen'? [-Wimplicit-function-declaration]
61 |         int bytes_sent = send(receiver, send_message, strlen(send_message), 0);
    |                                                         ^~~~~~
/usr/bin/ld: /tmp/ccJ0rPZ.o: in function 'forwardMessage':
chat_group.c:(.text+0x145): undefined reference to 'strlen'
collect2: error: ld returned 1 exit status
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_group.c -o build/chat_group && ./build/chat_group
Server is listening at port 8888...
New client accepted: 4
Client 4 is waiting
New client accepted: 5
Pair created: client 4 & client 5
New client accepted: 6
Client 6 is waiting
New client accepted: 7
Pair created: client 6 & client 7
```

- Khi client 1 ngắt kết nối, server sẽ đóng kết nối của client 2 và ngược lại

```

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Waiting for another client...
You are paired. Start chatting...
hello
Client 5: hi
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
You are paired. Start chatting...
xin chao
hiiii
Client 6: chao ban
^C
happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$

happycat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Waiting for another client...
You are paired. Start chatting...
Client 7: xin chao
Client 7: hiiii
chao ban

```

- Kết quả server

```

// chat_group.c
void *forwardMessage(void *arg) {
    while (1) {
        buffer[bytes_received] = '\0';

        // gui tin nhan vua nhan sang receiver
        char send_message[1048];
        snprintf(send_message, sizeof(send_message), "Client %d: %s", sender, buffer);
        int bytes_sent = send(receiver, send_message, strlen(send_message), 0);

        if (bytes_sent <= 0) {
            // ...
        }
    }
}

// chat_server.c
int main() {
    // ...
    while (1) {
        // ...
        if (bytes_received > 0) {
            // ...
            forwardMessage(&client);
        }
    }
}

```

```

happycat@LAPTOP-8SOA2IID: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_group.c -o build/chat_group && ./build/chat_group

Server is listening at port 8888...
New client accepted: 4
Client 4 is waiting
New client accepted: 5
Pair created: client 4 & client 5
New client accepted: 6
Client 6 is waiting
New client accepted: 7
Pair created: client 6 & client 7
Client 4 disconnected
Client 5 disconnected
Pair closed
Client 7 disconnected
Client 6 disconnected
Pair closed

```

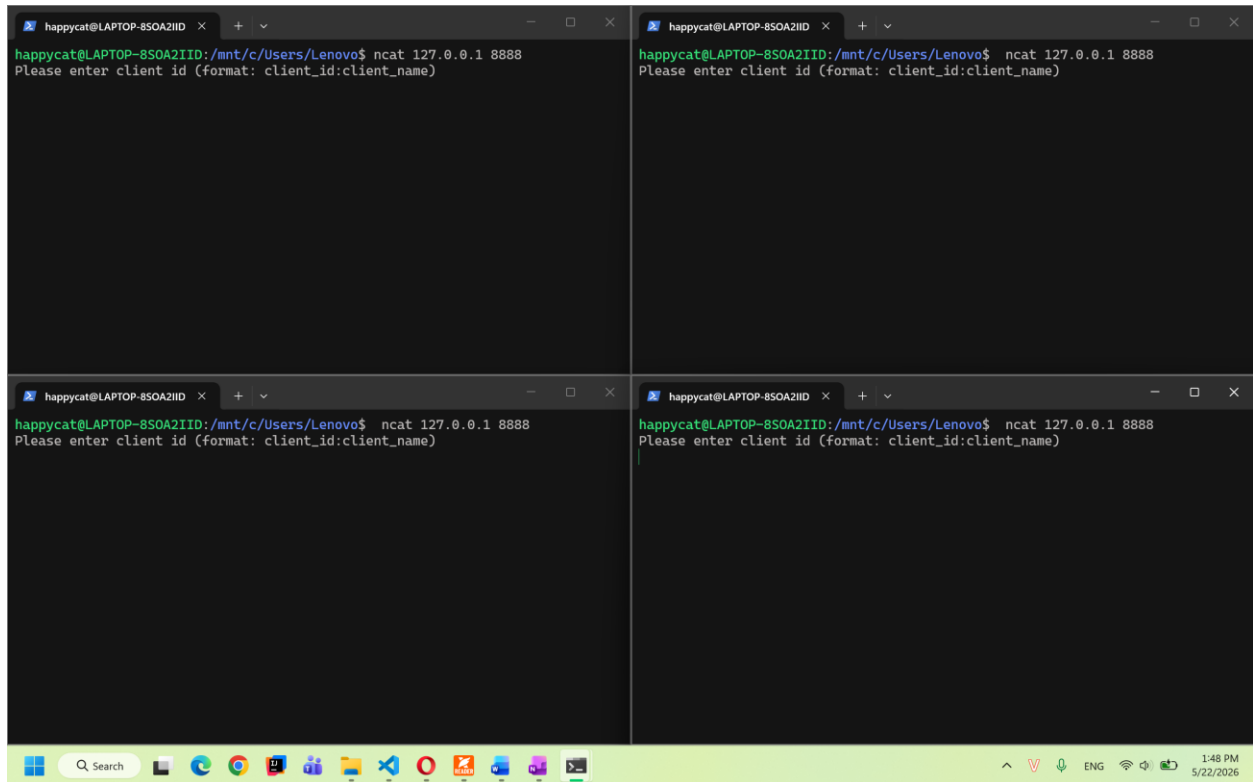

3. Bài 4.01 – chat_server
- Chạy server ở cổng 8888

The screenshot shows the Visual Studio Code editor with the file `chat_server_multithread.c` open. The code is written in C and includes a mutex to handle multiple clients. The terminal window shows the command to compile and run the server, and the output indicates that the server is listening on port 8888.

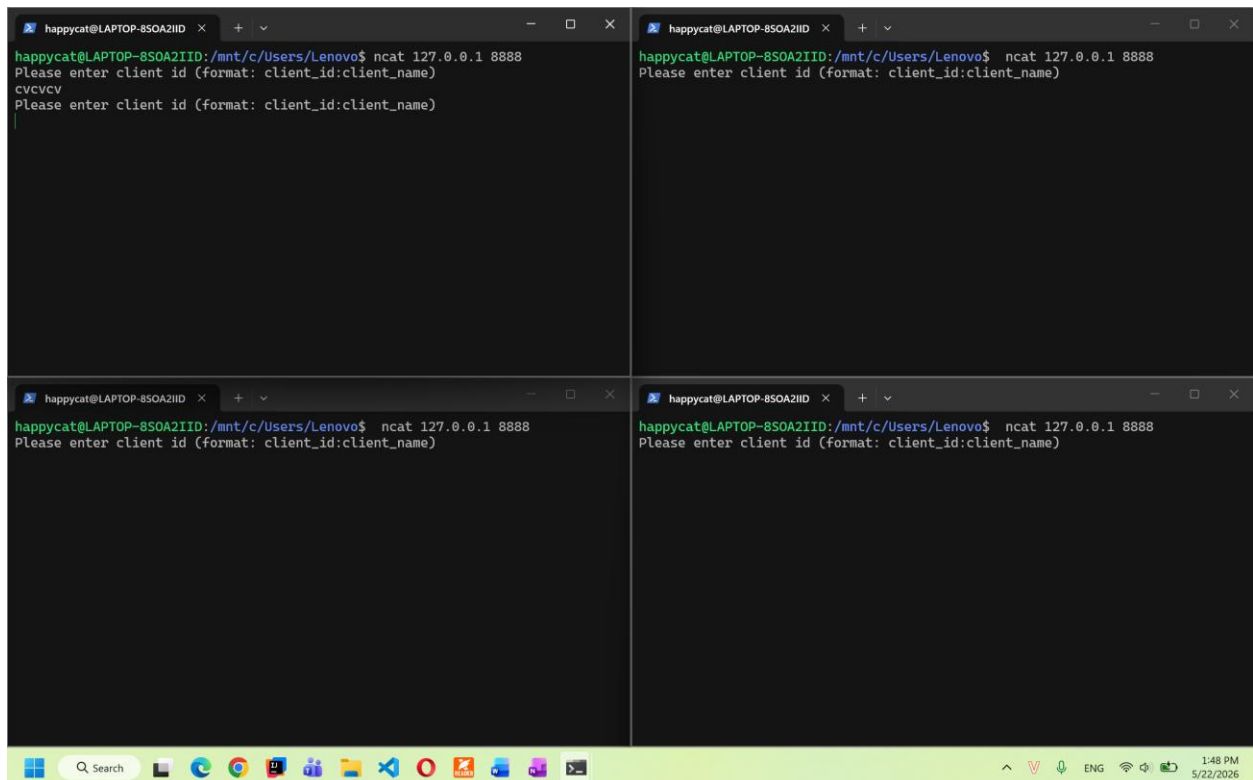
```
215
216     if (num_clients >= MAX_CLIENTS) {
217         pthread_mutex_unlock(&clients_mutex);
218
219         char full_msg[] = "Server is full\n";
220         send(client, full_msg, strlen(full_msg), 0);
221         close(client);
222         continue;
223     }
224
225     // chấp nhận và ghi vào mảng clients
226     clients[num_clients] = client;
```

```
happy@LAPTOP-850A2IID: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_server_multithread.c -o build/chat_server_multithread && ./build/chat_server_multithread
Server is listening at port 8888...
```

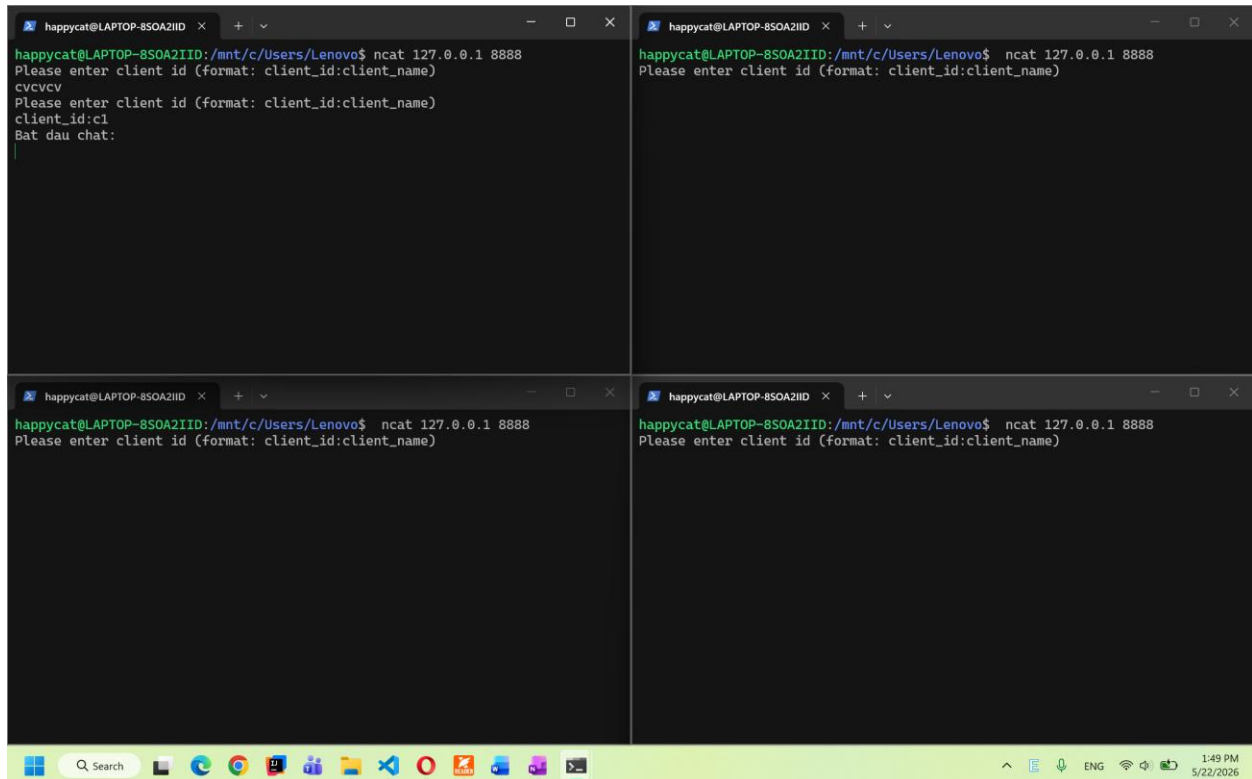
- Khi client kết nối, server gửi yêu cầu nhập client id



- Nếu client nhập sai, server sẽ gửi lại yêu cầu



- Client nhập đúng cú pháp thì server sẽ đánh dấu là đã nhập và gán tên cho client



The image shows four terminal windows arranged in a 2x2 grid. Each window is titled 'happy cat@LAPTOP-8SOA2IID'. The top-left window shows the server prompt 'Please enter client id (format: client_id:client_name)' and the user input 'cvcvcv'. The top-right window shows the same prompt and input. The bottom-left window shows the prompt and input 'client_id:c1'. The bottom-right window shows the prompt and input 'client_id:c1'. The system prompt 'Bat dau chat:' is visible in the bottom-left window. The Windows taskbar is visible at the bottom of the image.

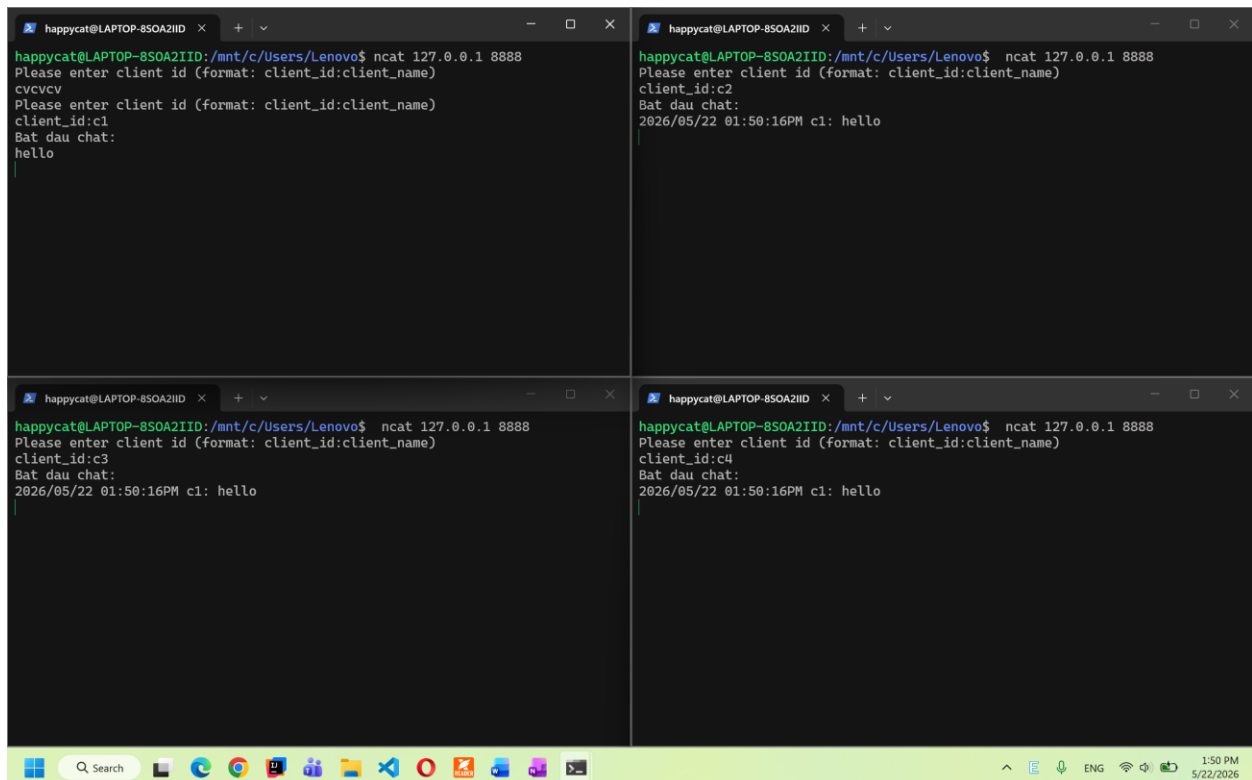
```
happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
cvcvcv
Please enter client id (format: client_id:client_name)
client_id:c1
Bat dau chat:

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
```

- Client 1 gửi thì các client còn lại sẽ nhận được dữ liệu và thời gian



The image shows four terminal windows arranged in a 2x2 grid. The top-left window shows the server prompt 'Please enter client id (format: client_id:client_name)' and the user input 'cvcvcv'. The top-right window shows the prompt and input 'client_id:c2'. The bottom-left window shows the prompt and input 'client_id:c3'. The bottom-right window shows the prompt and input 'client_id:c4'. The system prompt 'Bat dau chat:' is visible in the bottom-left window. The output '2026/05/22 01:50:16PM c1: hello' is visible in the bottom-left and bottom-right windows. The Windows taskbar is visible at the bottom of the image.

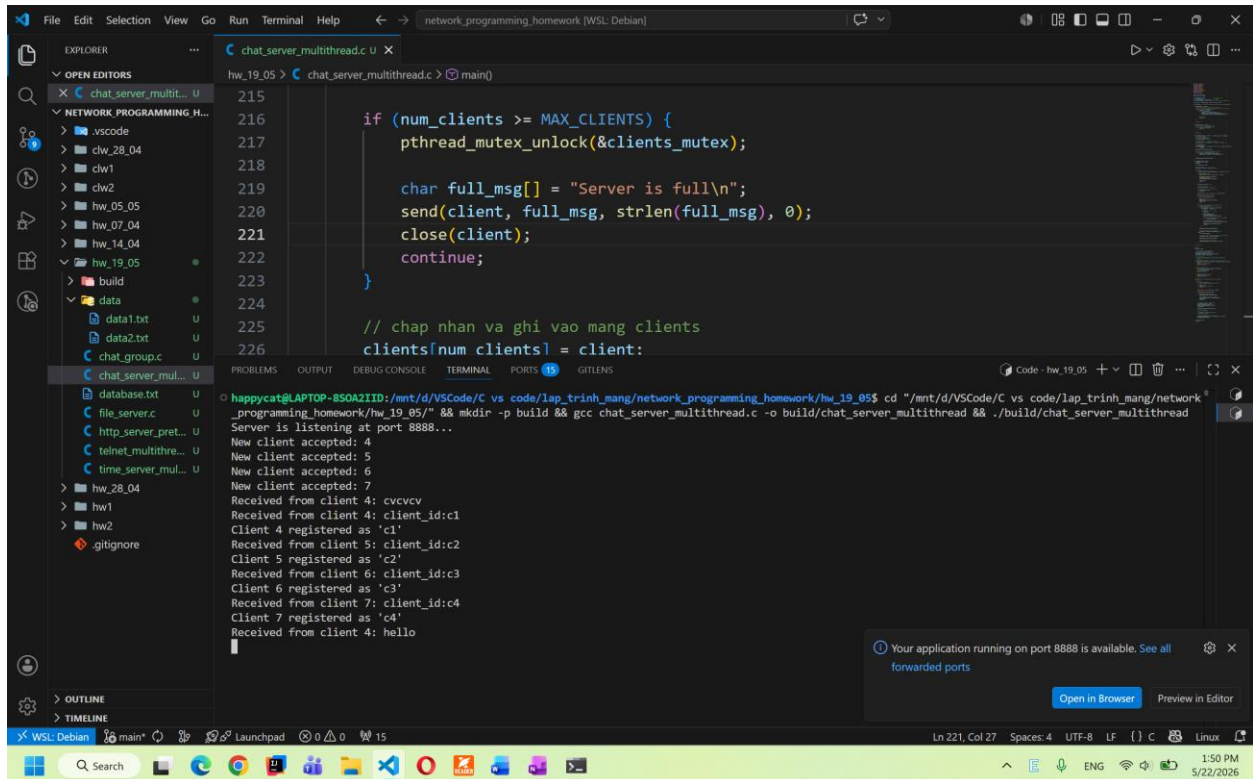
```
happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
cvcvcv
Please enter client id (format: client_id:client_name)
client_id:c1
Bat dau chat:
hello

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c2
Bat dau chat:
2026/05/22 01:50:16PM c1: hello

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c3
Bat dau chat:
2026/05/22 01:50:16PM c1: hello

happy cat@LAPTOP-8SOA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c4
Bat dau chat:
2026/05/22 01:50:16PM c1: hello
```

- Kết quả phía server



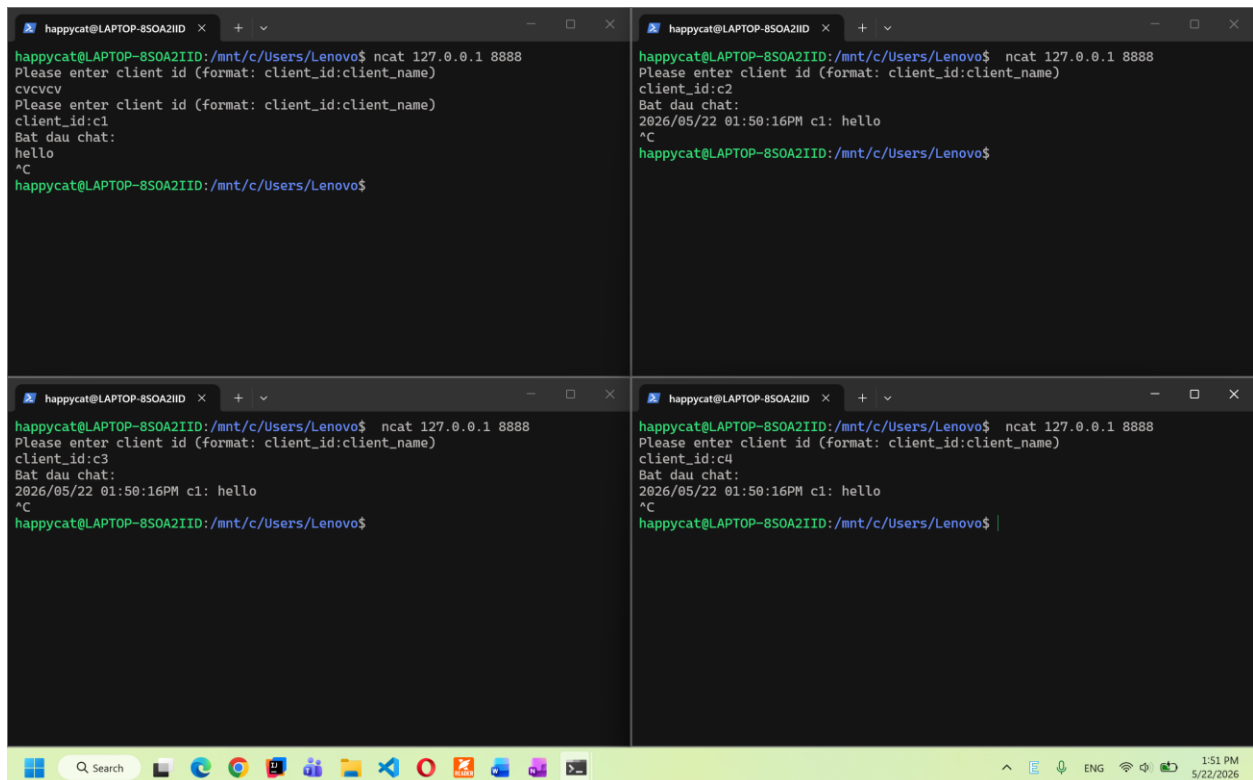
The screenshot shows the VS Code editor with the file `chat_server_multithread.c` open. The code is a C program for a multi-threaded chat server. It includes headers for `pthread` and `unistd.h`, defines `MAX_CLIENTS` as 10, and uses a mutex to manage client connections. The `main` function starts by listening on port 8888 and then enters a loop where it accepts new clients. If the number of clients reaches `MAX_CLIENTS`, it sends a "Server is full" message and closes the client. Otherwise, it registers the client and adds it to the `clients` array. The terminal output shows the server listening on port 8888 and accepting clients 4 through 7. Client 4 sends the message "hello".

```
hw_19_05 > chat_server_multithread.c (main)
215
216
217     if (num_clients >= MAX_CLIENTS) {
218         pthread_mutex_unlock(&clients_mutex);
219
220         char full_msg[] = "Server is full\n";
221         send(client, full_msg, strlen(full_msg), 0);
222         close(client);
223         continue;
224     }
225
226     // chap nhan va ghi vao mang clients
227     clients[num_clients] = client;
```

Terminal Output:

```
happyocat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc chat_server_multithread.c -o build/chat_server_multithread && ./build/chat_server_multithread
Server is listening at port 8888...
New client accepted: 4
New client accepted: 5
New client accepted: 6
New client accepted: 7
Received from client 4: cvcvcv
Client 4 registered as 'c1'
Received from client 5: client_id:c2
Client 5 registered as 'c2'
Received from client 6: client_id:c3
Client 6 registered as 'c3'
Received from client 7: client_id:c4
Client 7 registered as 'c4'
Received from client 4: hello
```

- Khi client ngắt kết nối



The screenshot shows four terminal windows, each representing a client connection to the chat server. Each window shows the client entering a client ID, sending a message, and then disconnecting (indicated by `^C`). The server responds with the client's message and then prompts for the next client ID.

Terminal 1 (Client 1):

```
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
cvcv
Please enter client id (format: client_id:client_name)
client_id:c1
Bat dau chat:
hello
^C
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

Terminal 2 (Client 2):

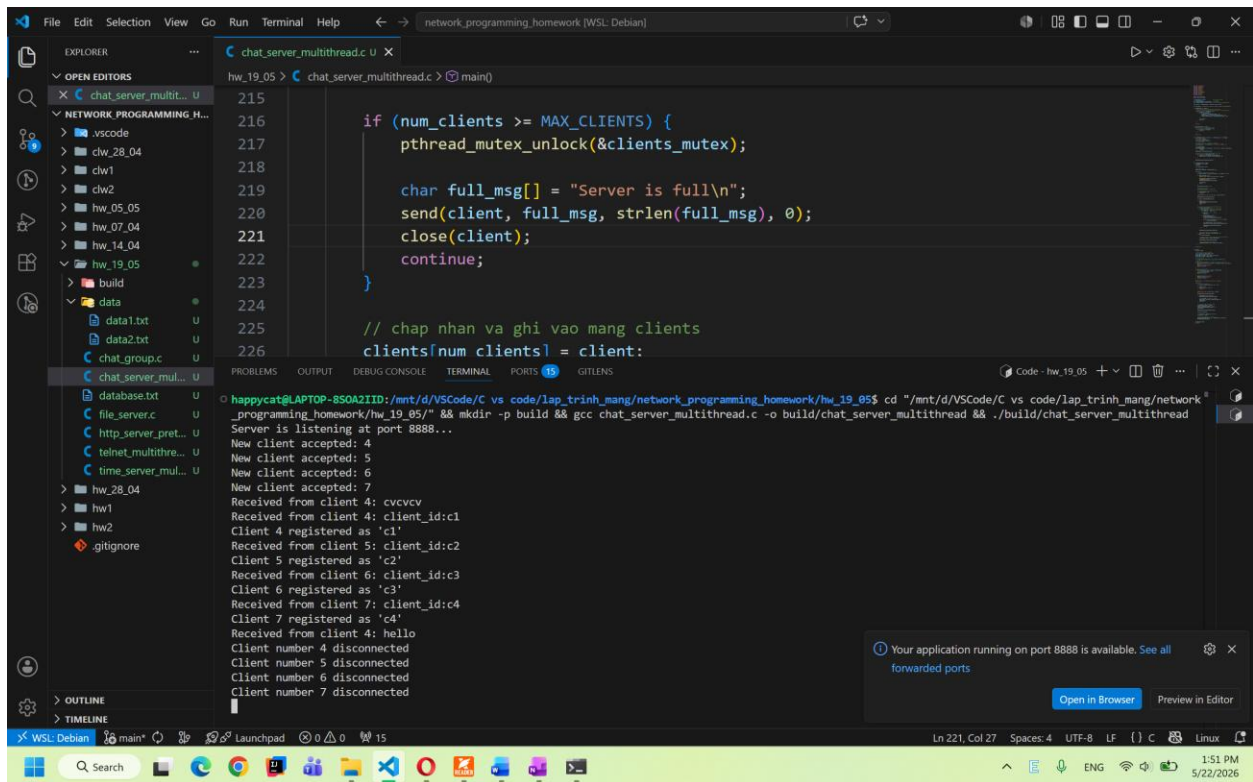
```
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c2
Bat dau chat:
2026/05/22 01:50:16PM c1: hello
^C
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

Terminal 3 (Client 3):

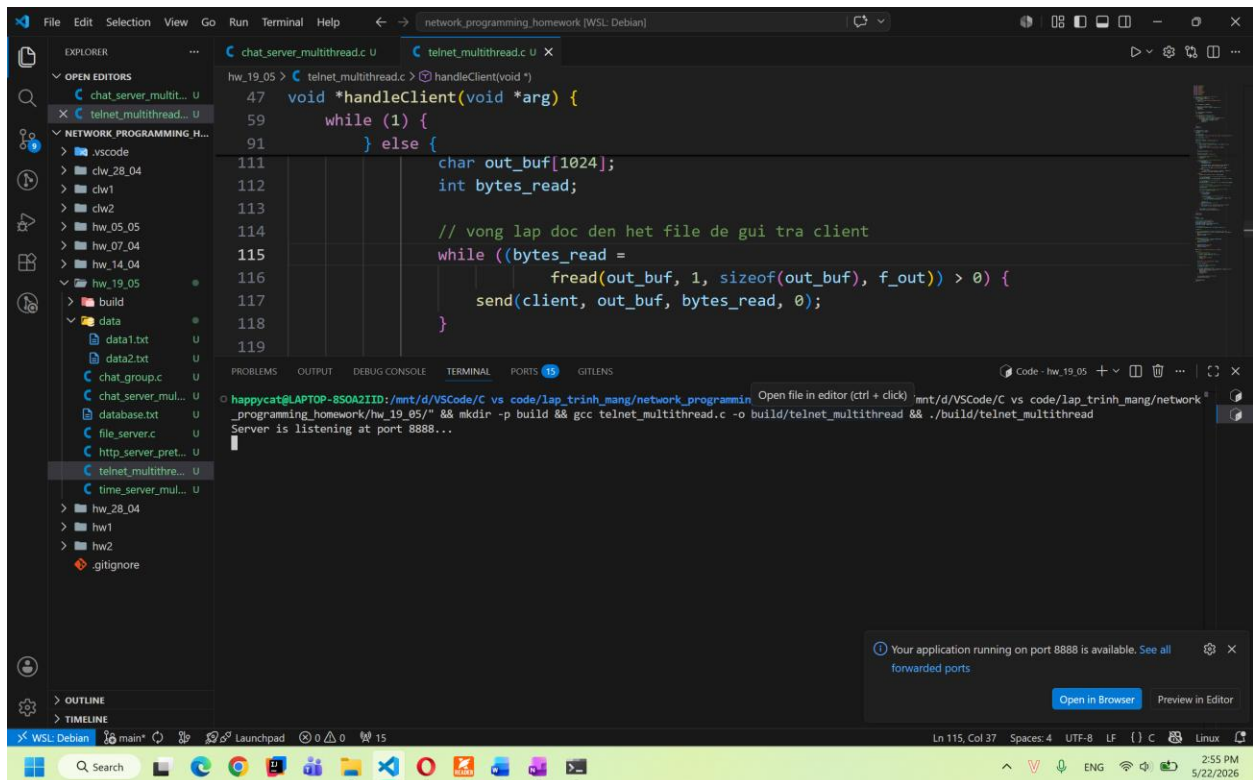
```
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c3
Bat dau chat:
2026/05/22 01:50:16PM c1: hello
^C
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

Terminal 4 (Client 4):

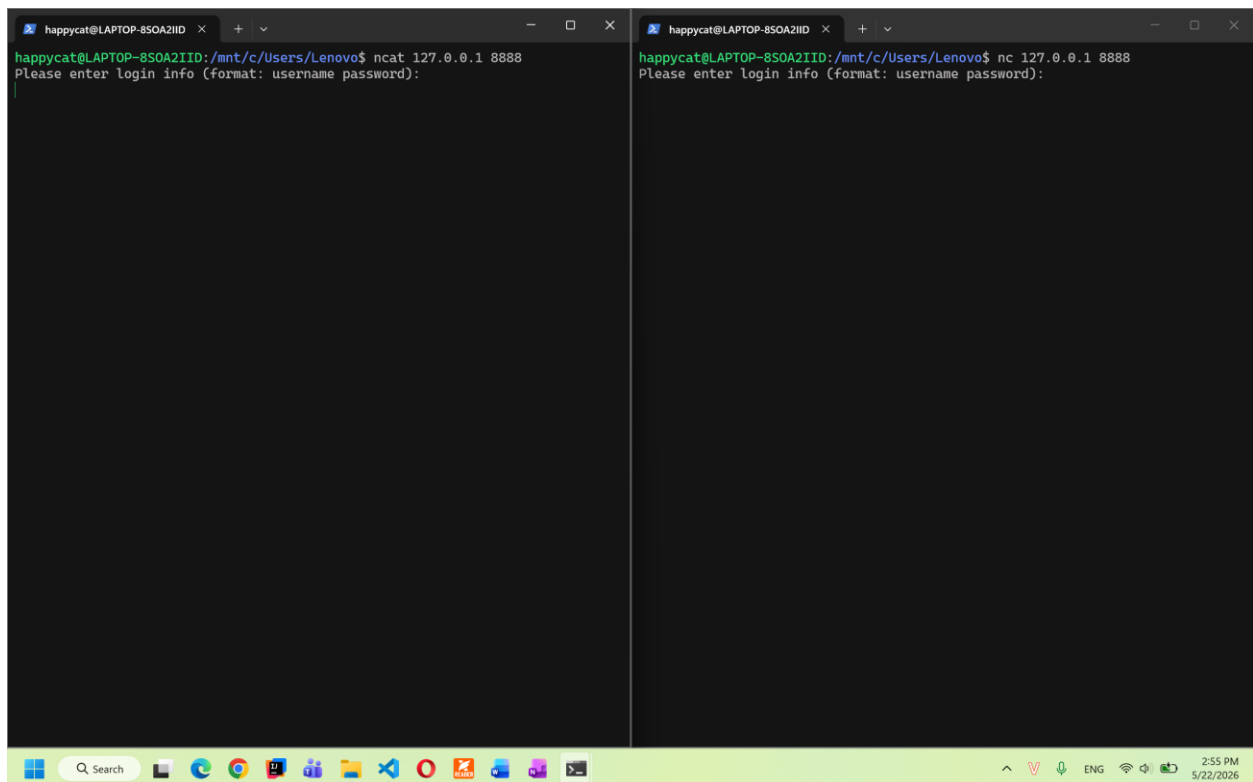
```
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c4
Bat dau chat:
2026/05/22 01:50:16PM c1: hello
^C
happyocat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```



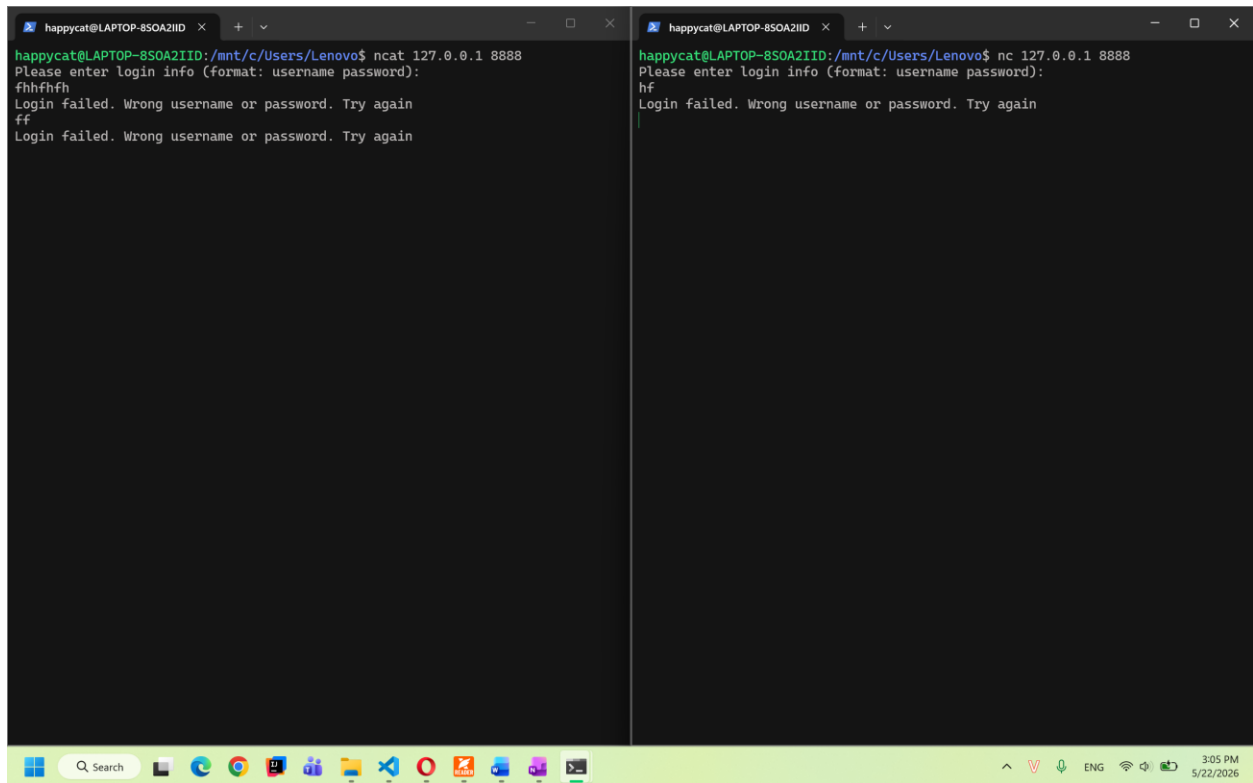
4. Bài 4.02 – telnet_server
 - Chạy server ở cổng 8888



- Chạy 2 client kết nối vào server



- Nhập username và password (format: “username password”)
- Nếu nhập sai, server sẽ báo lỗi và yêu cầu nhập lại

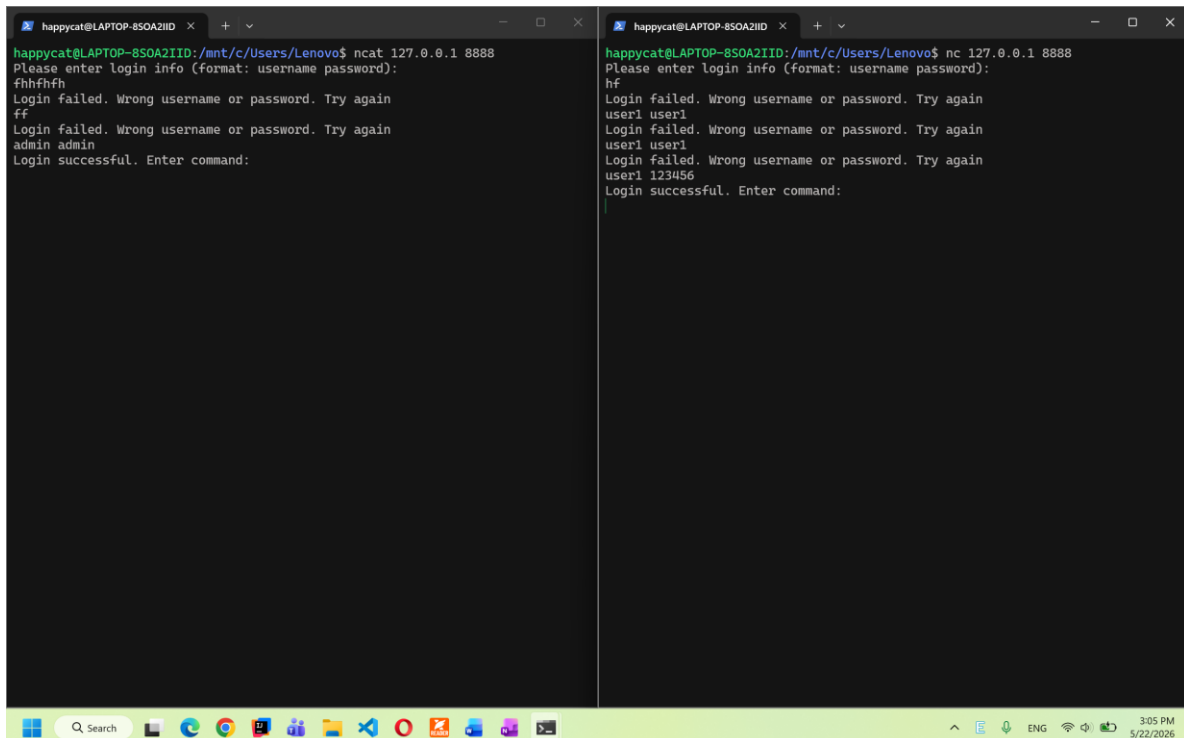


The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-85QA2IID'. The left terminal shows a user running 'ncat 127.0.0.1 8888'. The prompt is 'Please enter login info (format: username password):'. The user enters 'fhhfhfh', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'ff', and the server responds with 'Login failed. Wrong username or password. Try again'. The right terminal shows a user running 'nc 127.0.0.1 8888'. The prompt is 'Please enter login info (format: username password):'. The user enters 'hf', and the server responds with 'Login failed. Wrong username or password. Try again'. The Windows taskbar is visible at the bottom, showing the time as 3:05 PM on 5/22/2026.

```
happycat@LAPTOP-85QA2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
fhhfhfh
Login failed. Wrong username or password. Try again
ff
Login failed. Wrong username or password. Try again
```

```
happycat@LAPTOP-85QA2IID:/mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
hf
Login failed. Wrong username or password. Try again
```

- Nhập đúng thì server sẽ báo thành công và yêu cầu nhập lệnh



The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-85QA2IID'. The left terminal shows a user running 'ncat 127.0.0.1 8888'. The prompt is 'Please enter login info (format: username password):'. The user enters 'fhhfhfh', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'ff', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'admin admin', and the server responds with 'Login successful. Enter command:'. The right terminal shows a user running 'nc 127.0.0.1 8888'. The prompt is 'Please enter login info (format: username password):'. The user enters 'hf', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'user1 user1', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'user1 user1', and the server responds with 'Login failed. Wrong username or password. Try again'. The user enters 'user1 123456', and the server responds with 'Login successful. Enter command:'. The Windows taskbar is visible at the bottom, showing the time as 3:05 PM on 5/22/2026.

```
happycat@LAPTOP-85QA2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
fhhfhfh
Login failed. Wrong username or password. Try again
ff
Login failed. Wrong username or password. Try again
admin admin
Login successful. Enter command:
```

```
happycat@LAPTOP-85QA2IID:/mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
hf
Login failed. Wrong username or password. Try again
user1 user1
Login failed. Wrong username or password. Try again
user1 user1
Login failed. Wrong username or password. Try again
user1 123456
Login successful. Enter command:
```


- Nhập lệnh xong, server sẽ thực hiện, ghi kết quả ra file out_{pid}.txt (sau khi dùng xong file này sẽ được xóa để tránh xung đột) và gửi về client

```

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
fhhfhfh
Login failed. Wrong username or password. Try again
ff
Login failed. Wrong username or password. Try again
admin admin
Login successful. Enter command:
ls
build
chat_group.c
chat_server_multithread.c
data
database.txt
file_server.c
http_server_prethreading.c
out_4.txt
telnet_multithread.c
time_server_multithread.c

Enter command: ls
build
chat_group.c
chat_server_multithread.c
data
database.txt
file_server.c
http_server_prethreading.c
out_4.txt
telnet_multithread.c
time_server_multithread.c

Enter command:

```

```

happycat@LAPTOP-85QA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Please enter login info (format: username password):
hf
Login failed. Wrong username or password. Try again
user1 user1
Login failed. Wrong username or password. Try again
user1 user1
Login failed. Wrong username or password. Try again
user1 123456
Login successful. Enter command:
pwd
/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05

Enter command: |

```

- Kết quả ở server

```

// chat_server_multithread.c
47 void *handleClient(void *arg) {
59     while (1) {
91     } else {
92         printf("Client %d execute: %s\n", client, buffer);
93
94         // moi client su dung 1 file rieng de tranh canh tranh
95         char out_filename[64];
96         snprintf(out_filename, sizeof(out_filename), "out_%d.txt", client);
97
98         // tao cau lenh: "lenh > out_{client}.txt"
99         char sys_cmd[1280];
100        snprintf(sys_cmd, sizeof(sys_cmd), "%s > %s", buffer, out_filename);

```

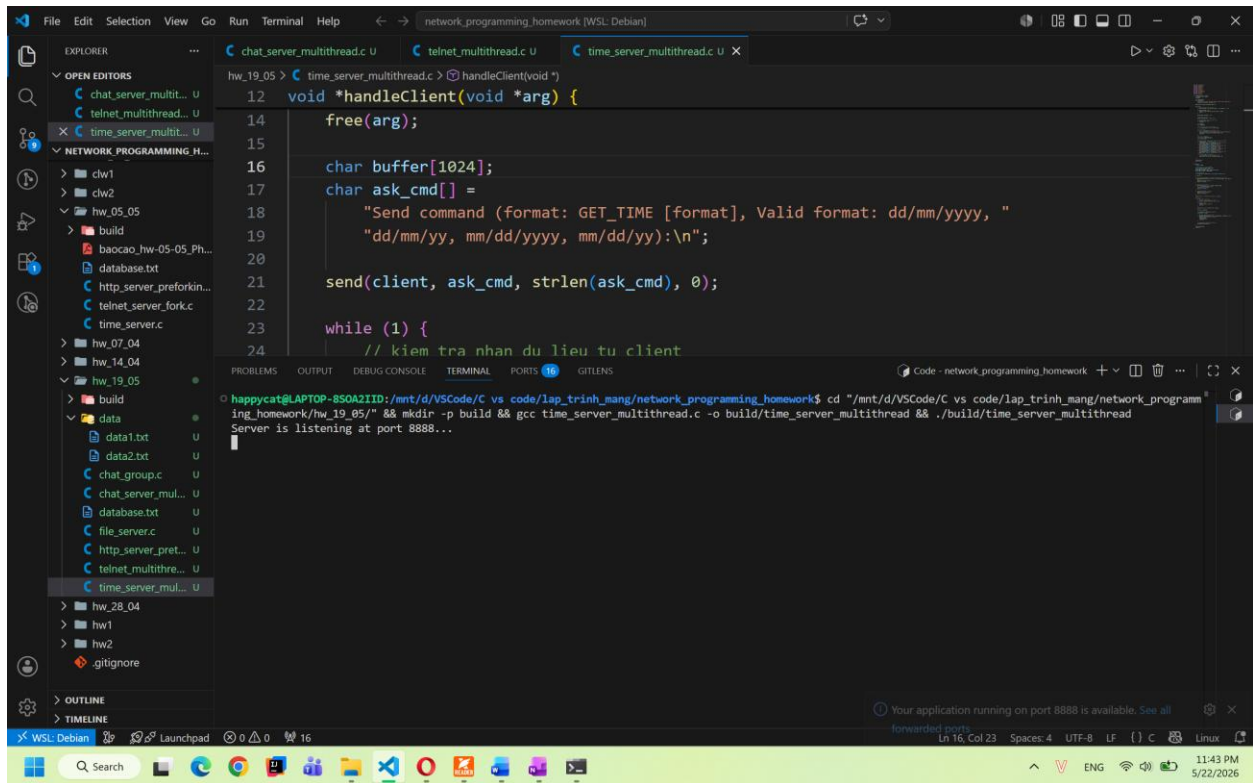
```

happycat@LAPTOP-85QA2IID: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc telnet_multithread.c -o build/telnet_multithread && ./build/telnet_multithread
Server is listening at port 8888...
New client accepted: 4
New client accepted: 5
Client 4 logged in successfully
Client 5 logged in successfully
Client 4 execute: ls
Client 4 execute: ls
Client 5 execute: pwd

```


5. Bài 4.03 – time_server

- Chạy server ở cổng 8888

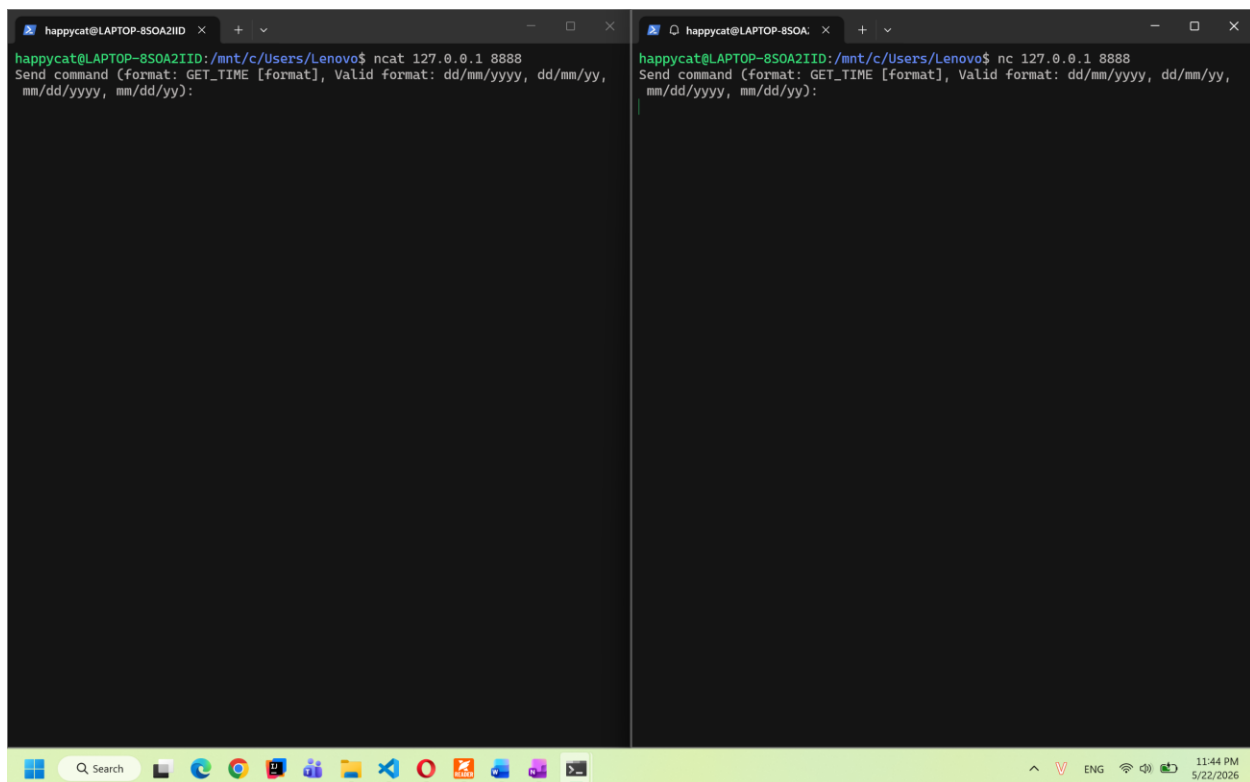


```
hw_19_05 > C time_server_multithread.c > handleClient(void *)
12 void *handleClient(void *arg) {
13
14     free(arg);
15
16     char buffer[1024];
17     char ask_cmd[] =
18         "Send command (format: GET_TIME [format], Valid format: dd/mm/yyyy, \"
19         \"dd/mm/yy, mm/dd/yyyy, mm/dd/yy):\\n\";
20
21     send(client, ask_cmd, strlen(ask_cmd), 0);
22
23     while (1) {
24         // kiểm tra nhận dữ liệu từ client
25     }
```

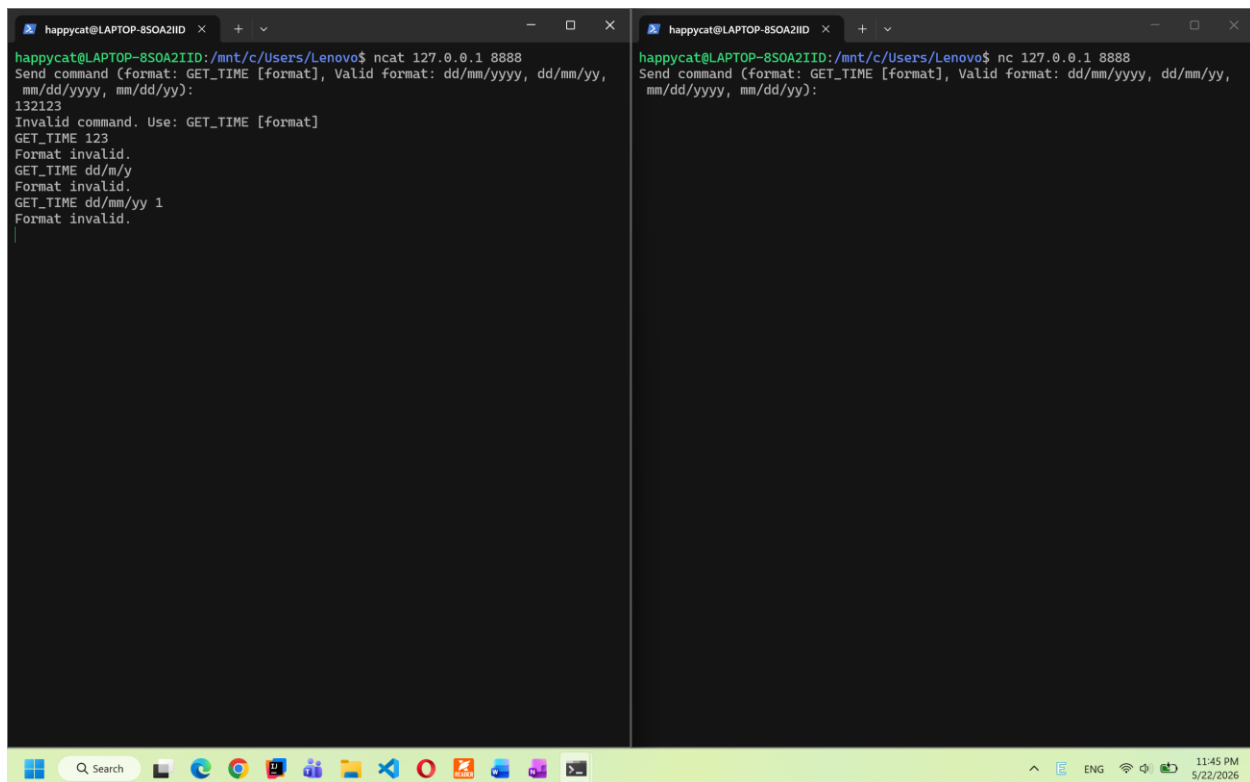
```
happyat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/$ cd "/mnt/d/VSCode/C" && gcc time_server_multithread.c -o build/time_server_multithread && ./build/time_server_multithread
Server is listening at port 8888...
```

Your application running on port 8888 is available. See all forwarded ports

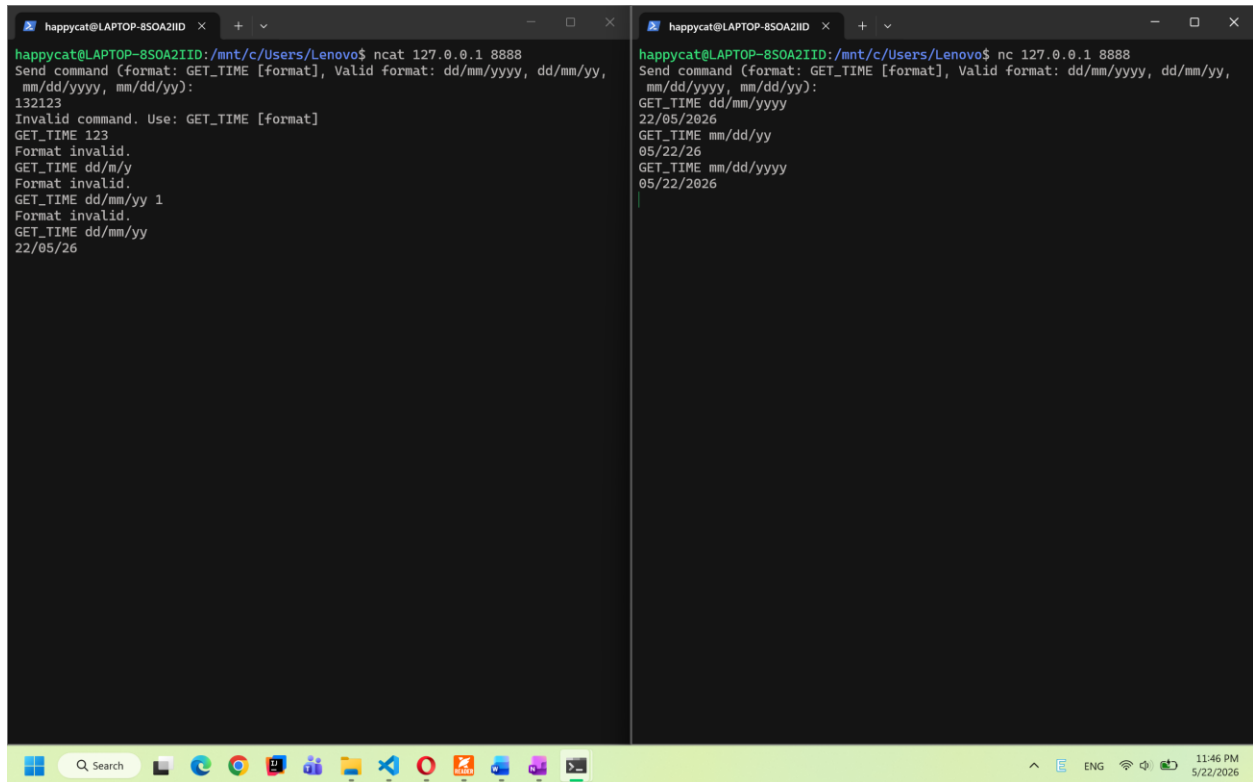
- Tạo 2 client kết nối với server



- Nếu nhập sai cú pháp, server sẽ yêu cầu nhập lại (thừa tham số, sai định dạng)



- Nhập đúng định dạng thì server sẽ gửi thời gian cho client

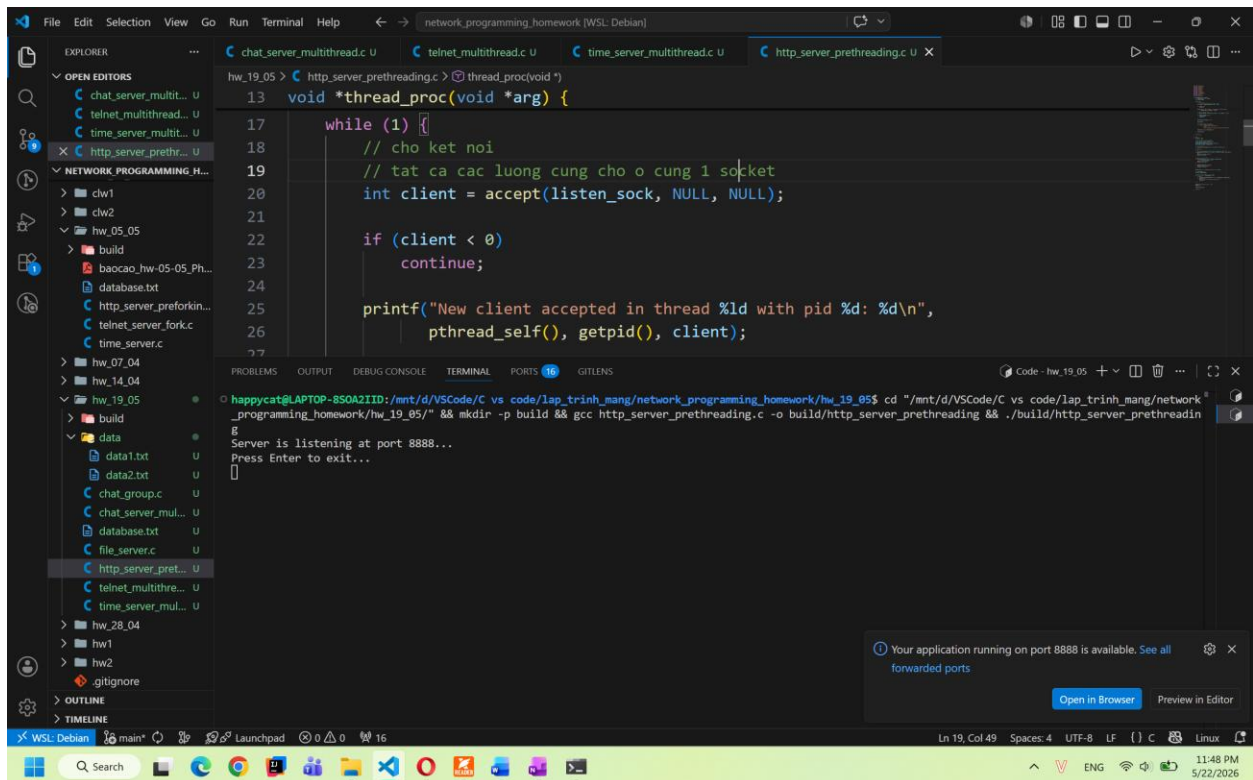


```
happycat@LAPTOP-850A2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Send command (format: GET_TIME [format], Valid format: dd/mm/yyyy, dd/mm/yy,
mm/dd/yyyy, mm/dd/yy):
132123
Invalid command. Use: GET_TIME [format]
GET_TIME 123
Format invalid.
GET_TIME dd/m/y
Format invalid.
GET_TIME dd/mm/yy 1
Format invalid.
GET_TIME dd/mm/yy
22/05/26

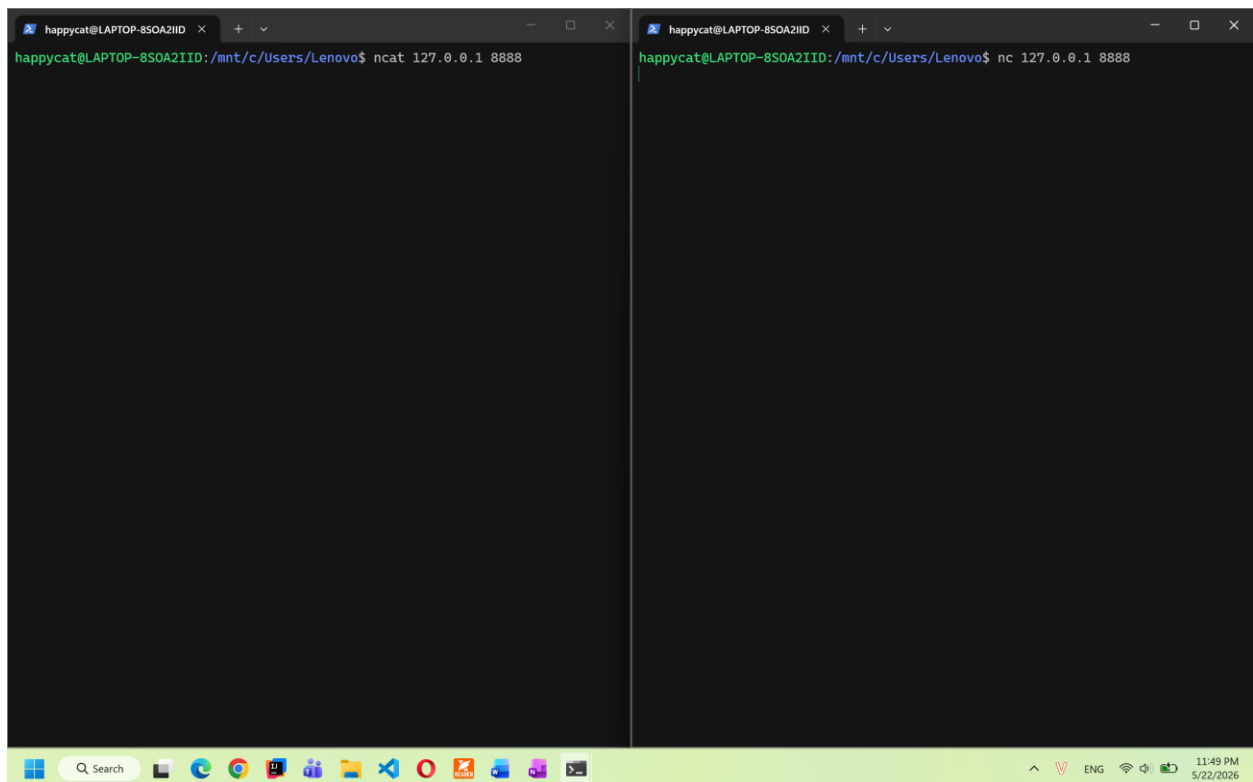
happycat@LAPTOP-850A2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
Send command (format: GET_TIME [format], Valid format: dd/mm/yyyy, dd/mm/yy,
mm/dd/yyyy, mm/dd/yy):
GET_TIME dd/mm/yyyy
22/05/2026
GET_TIME mm/dd/yy
05/22/26
GET_TIME mm/dd/yyyy
05/22/2026
```

6. Bài 4.04 – http_server

- Chạy server ở cổng 8888



- Tạo 2 client kết nối với server



- Client gửi message và nhận được mã HTML từ server, sau đó server đóng kết nối với client vừa gửi

```

happyocat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
123
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xin chao cac ban</h1></body></html>

happyocat@LAPTOP-8SOA2IID: /mnt/c/Users/Lenovo$ nc 127.0.0.1 8888
456
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xin chao cac ban</h1></body></html>

```

- Bên phía server, khi nhấn Enter sẽ kill tất cả tiến trình

```

// http_server_prethreading.c
13 void *thread_proc(void *arg) {
14     while (1) {
15         // cho ket noi
16         // tat ca cac luong cung cho o cung 1 socket
17         int client = accept(listen_sock, NULL, NULL);
18
19         if (client < 0)
20             continue;
21
22         printf("New client accepted in thread %ld with pid %d: %d\n",
23               pthread_self(), getpid(), client);
24     }
25 }
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
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81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100

```

```

happyocat@LAPTOP-8SOA2IID: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_19_05/" && mkdir -p build && gcc http_server_prethreading.c -o build/http_server_prethreading && ./build/http_server_prethreading
Server is listening at port 8888...
Press Enter to exit...
New client accepted in thread 132159548434624 with pid 181854: 4
New client accepted in thread 132159532041928 with pid 181854: 5
123
456

```